

Exactly Two Seeds

A Constraint-Derived Classification of the Digit-Extraction Towers, with Predicted Members and Inverse Recovery

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ABSTRACT

We ask whether a constraint-first construction can produce a distinguished mathematical object without being supplied that object in advance, and we answer it constructively for a family containing the Bailey–Borwein–Plouffe formula. Two independently motivated conditions — a pole-annihilation requirement and a zero-sum requirement — are imposed simultaneously on a coefficient space. Neither selects anything alone; jointly they select a unique line, whose weights are $[4, -2, -1, -1]$. We show this signature is a consequence of the single condition $\operatorname{Re}(z_{\text{char}}) = 1$, which also forces the radix $b = \sec^m(2\pi/m)$ and, through the annihilator's degree, the shell support $\{1, m-4, m-3, m-2\}$. The construction therefore has a closed generator, $4S(1) - 2S(m-4) - S(m-3) - S(m-2)$, of which the classical base-16 formula for π is the $m = 8$ instance. We then prove a classification: $\sec^m(2\pi/m)$ is a positive integer greater than one for exactly two values, $m = 6$ and $m = 8$, the argument turning on the fact that $\mathbb{Q}(2\cos(2\pi/m))$ is abelian while $\mathbb{Q}(2^{1/e})$ is not for $e \geq 3$. Non-primitive indices are lifts, $n = km$ with $b = b_{\text{seed}}^k$, so that the apparent proliferation of towers is an identity rather than a phenomenon and the index n is not a tower's address — the pair (m, k) is. Four previously unbuilt towers predicted this way were constructed and verified, and an inverse map recovers $(m, k, \theta, b_{\text{seed}})$ from an observed tower, round-tripping all eight cases exactly. Negative controls confirm the method returns *no closure* under dependence, over-constraint, under-constraint, and single-channel selection. The same closure architecture is exhibited in three further substrates — a Boolean cryptographic round, a driven nonlinear oscillator, and a quantum spectral problem — as evidence that the reading operation is not an artifact of linear algebra.

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

1. A constant that is nowhere

Take a three-digit number. Reverse it. Subtract the smaller from the larger. Reverse that difference and add. The answer is 1089. It is always 1089. It does not matter which number you started with.

The interesting question is not why the answer is constant. It is *where the constant lives*. It is not in the number you chose, because any number works. It is not in the subtraction, which alone produces nine different answers. It is not in the reverse-and-add, which alone produces nothing until it is handed a difference. The 1089 is not stored anywhere in the procedure. It appears where two operations, each destroying what the other preserves, are composed.

Write the number as $100a + 10b + c$ and its reversal as $100c + 10b + a$. Subtracting gives $99 \cdot (a - c)$: the middle digit is gone, because reversal is a reflection and b sits on the mirror's axis, where self minus self is zero. Three free parameters have become one. The difference is $99k$ for a span k between 1 and 9, which as three digits reads $(k-1, 9, 10-k)$; reversing and adding pairs the outer digits as $(k-1) + (10-k) = 9$ and doubles the middle to 18, giving $900 + 180 + 9 = 1089$ with k cancelled. The second operation annihilates exactly what the first preserved.

So the constant depends on neither free parameter. It is not a property of the starting number, nor of either operation alone. It exists only in the composition, and it is precisely what survives having everything else removed. This paper takes that structure and asks whether a well-known object in analysis has the same status — and then asks what else the structure generates once it is stated exactly.

2. The closure, and what it must be able to refuse

Name the two operations by what they do. The first fixes what can be distinguished; call it C_0 . The second fixes what change is permitted; call it C_1 . These are structural roles, not physical categories: in one system C_0 is a pole geometry and C_1 an algebraic side-condition, in another C_0 is a fixed law and C_1 an applied drive, and in a third C_1 is an admissibility requirement with no drive anywhere. What is invariant is the pairing, not the labels.

Neither has a distinguished output alone. The condition of interest is the joint one:

$$\mathcal{M}(C_0, C_1; x) = 0$$

What makes this more than "solve both equations" is that the joint condition can be decisive where neither separate condition is. A structure selected this way we call a **closure**, and it has a testable signature: remove either constraint and the selection is destroyed — not weakened, destroyed, since the candidate space reopens and nothing in it is distinguished.

2.1 Simultaneity

The two must act at once on the same candidate. Applied in sequence — satisfy C_0 , freeze the result, then check C_1 — whatever C_0 left undetermined is baked into the intermediate object, and C_1 can no longer separate that residue from structure. This is the principle of a balanced signal line: the receiver does not compare two copies for agreement, it takes their difference, and interference common to both conductors cancels only because the subtraction is simultaneous.

2.2 Two incomplete readings

The test is not a conjunction. If "incomplete" means a residual deviation remains, two readings with equal and opposite residuals sum to zero: each is individually wrong and jointly they are exact. The acceptance condition is $r_{\text{forward}} + r_{\text{return}} = 0$. A conjunction cannot express this, having access only to whether failures occurred and not to their magnitudes; two failures that annihilate are indistinguishable, under conjunction, from two that compound. Section 5 gives a concrete instance in which one construction fixes a numerator's structure but not its scale, and the scale is determined only by closing two constructions against each other.

2.3 Negative controls

A method that never refuses is not a method. Working in the six-dimensional coefficient space of Section 3, the true case gives admissible dimensions 2 under C_0 alone, 5 under C_1 alone, and 1 under both, with the reductions adding exactly — four dimensions removed plus one, five removed jointly, no overlap. Four controls were run against this baseline.

Control	Construction	Dimensions (C_0 / C_1 / joint)	Verdict
— (true case)	pole geometry $\times$ zero-sum	2 / 5 / 1	closure
A. dependent	$C_0 \times$ a row of itself	2 / 5 / 2	no closure — overlapping
B. over-constrained	full closure $\times$ two more conditions	1 / 4 / 0	no closure — empty
C. under-constrained	two weak single conditions	5 / 5 / 4	no closure — still 4-dim
D. single-channel	A selects alone; B already satisfied	1 / 5 / 1	no closure — selection survives

Control D is the important one and deserves care, because it is the case that most resembles success. There the joint set *is* one-dimensional, so the construction looks like a closure by dimension count alone. But removing C_1 leaves it one-dimensional still: the selection was A's property all along, and the vanishing test rejects it. The distinction the controls enforce is that **joint satisfaction is not joint selection**.

3. The construction

For a denominator $P(t) = b - t^n$ with $b > 1$, define the shell sums

$$S(r) = \sum_{k \geq 0} 1 / (b^k (nk + r)) = b \int_0^1 t^{r-1} / P(t) dt, \quad r = 1 \dots n.$$

A digit-extraction formula is a rational combination $\sum_r c_r S(r)$ evaluating to a target constant, and the coefficient vector is the object under constraint. Each $S(r)$ is a rational combination of finitely many transcendental atoms — logarithms and π -type terms — so the passage from coefficient vectors to atoms is a rational linear operator Φ . The two constraints are conditions on Φ , and neither mentions a target vector.

- **C₀ — pole geometry.** The numerator must annihilate every pole not at the characteristic angle, so that $\Phi(c)$ has vanishing projection onto all non- π atoms.
- **C₁ — zero-sum.** The coefficients must sum to zero: the instrument has no response to a constant.

3.1 A worked case, and the vanishing

Take $b = 64$, $n = 6$, where $64 - t^6 = -(t - 2)(t + 2)(t^2 - 2t + 4)(t^2 + 2t + 4)$. The six roots are the sixth roots of 64 on the circle of radius 2, and $\rho = 2$ satisfies $\rho^6 = 64$. Reducing the six shell integrals to the basis $\{\log 2, \log 3, \log 7, \sqrt{3} \cdot \pi, \sqrt{3} \cdot \arctan(\sqrt{3}/2)\}$ gives a 6×5 matrix of rank 5, with one-dimensional kernel $z = (16, -24, -8, -6, 1, 0)$, $\sum z = -21$. Imposing C_0 alone leaves a two-dimensional space; C_1 alone leaves five; both together leave a unique line, which scaled to integers is

$$(36, -18, -9, -9, 0, 0) = 9 \cdot (4, -2, -1, -1, 0, 0).$$

The active weights are $9 \cdot [4, -2, -1, -1]$ — the same signature the classical base-16 formula for π carries. It was not supplied; the computation was given two constraints and no target. Two facts sharpen this. The vector $(4, -2, -1, -1, 0, 0)$ lies *inside* both admissible spaces and is distinguished in neither, since each has dimension above one — membership is not selection. And the constraints are strictly independent, removing four and one dimension respectively and five jointly, with no overlap.

4. The signature is derived before any tower is named

A shared vector across two constructions invites the objection that the second was built to match the first. The objection is answered by deriving the signature from a condition mentioning neither. The condition is that the real part of the surviving characteristic pole is one:

$$\text{Re}(z_{\text{char}}) = 1$$

This is what makes the surviving pole contribute to π through an arctangent-type integral whose argument structure requires unit real part. From it, three things follow by elementary algebra.

First the radix. $\text{Re}(z_{\text{char}}) = 1$ means $\rho \cdot \cos\theta = 1$, so $b = (1/\cos\theta)^n = \sec^n\theta$. For $(n, \theta) = (6, 60^\circ)$ this is $2^6 = 64$; for $(8, 45^\circ)$ it is $(\sqrt{2})^8 = 16$. The bases are outputs, not choices.

Second the annihilator's sub-leading coefficient. The n roots sum to zero; the characteristic pair contributes $2 \cdot \text{Re}(z_{\text{char}}) = 2$; the real roots contribute zero, occurring in $\pm$ pairs. So the annihilated roots sum to -2 and the sub-leading coefficient is 2 in every member.

Third the constant term. Each annihilated conjugate pair contributes $|\text{root}|^2 = \rho^2$, and $\rho^{2d} = b^{2d/n} = 4$ in both towers, since $64^{1/3} = 16^{1/2} = 4$. So every annihilator in the class has descending coefficients beginning $[1, 2, 4]$, and the zero-sum multiplier $(1 - t)$ takes successive differences of those, yielding the non-zero pattern $[4, -2, -1, -1]$. No step is observation.

5. Two incomplete constructions, one closed ledger

Each construction alone leaves exactly one thing undetermined. Writing the numerator as $N(t) = c \cdot (1 - t) \cdot A(t)$, the annihilator A is fixed by C_0 and the factor $(1 - t)$ by C_1 , but the scalar c is fixed by neither. The base-16 construction has $c = 1$ and the base-64 construction has $c = 9/4$, and neither can say why. Closing them against one another determines it exactly:

$$\sqrt{3} = (64/16) \cdot (9/4) \cdot (\sqrt{3}/9)$$

The three factors come from three independent sources: 4 from the bases, which are themselves forced; $\sqrt{3}/9$ from the ratio of the bare integrals, fixed by pole geometry; and $9/4$, the residual. The identity holds if and only if the scale ratio is $\sqrt{3} \div (4 \cdot \sqrt{3}/9) = 9/4$, which is what it is. This is the fourth row of §2.2 realized: the forward construction is incomplete, the return relation is incomplete, neither is a proof, and their residuals cancel.

6. The seed theorem

The radix is now a function of the period alone, so the question of which constructions exist becomes a question in elementary number theory. Write $\theta = 2\pi/m$ for the primitive angle and ask for which m the radix is an admissible integer.

Theorem. $\sec^m(2\pi/m)$ is an integer $> 1 \Leftrightarrow m \in \{6, 8\}$.

Proof. Let $\beta = 2\cos(2\pi/m) = \zeta + \zeta^{-1}$ with $\zeta = e^{2\pi i/m}$. As a sum of roots of unity, β is an algebraic integer. Since $b = \sec^m(2\pi/m) = 2^m/\beta^m$, requiring $b \in \mathbb{Z}$ makes $\beta^m = 2^m/b$ rational; an algebraic integer that is rational is a rational integer, and it divides 2^m , so $\beta^m = 2^j$ for some integer j . For $m \geq 5$ we have $\cos(2\pi/m) > 0$, hence $\beta > 0$ and $\beta = 2^{j/m}$.

Write $j/m = p/e$ in lowest terms, so $\beta = 2^{p/e}$ and $\mathbb{Q}(\beta) = \mathbb{Q}(2^{1/e})$. Now $\mathbb{Q}(\beta)$ is the maximal real subfield of the cyclotomic field $\mathbb{Q}(\zeta_m)$; cyclotomic extensions are abelian and every subfield of an abelian extension is abelian, so $\mathbb{Q}(\beta)$ is abelian over $\mathbb{Q}$. But for $e \geq 3$, $x^e - 2$ is irreducible by Eisenstein and $\mathbb{Q}(2^{1/e})$ is not normal over $\mathbb{Q}$ — the real radical field omits the non-real conjugates $2^{1/e}\zeta_e^r$ — hence not abelian. Therefore $e \leq 2$.

If $e = 1$ then β is a positive rational integer with $0 < \beta < 2$ for $m \geq 5$, so $\beta = 1$, giving $\cos(2\pi/m) = 1/2$ and $m = 6$, whence $b = 64$. If $e = 2$ then $2j/m$ is an integer and β^2 is a power of two with $0 < \beta^2 < 4$, so $\beta^2 = 2$, giving $\cos(2\pi/m) = \sqrt{2}/2$ and $m = 8$, whence $b = 16$. ■

The boundary deserves an explicit remark rather than a dismissal. At $m = 3$ the value $\sec^3(120^\circ) = -8$ is an integer; it is excluded by positivity, not by integrality, and the theorem is a statement about admissible radices $b > 1$. At $m = 4$ the secant is undefined. A numerical scan over $m = 3 \dots 4000$ returns exactly the two predicted seeds, and the scan is exhaustive rather than truncated because $\sec^m(2\pi/m)$ decreases toward 1 as m grows, so no large- m case lies above any cutoff.

7. Lifts, and why n is not an address

A characteristic angle need not be indexed primitively. If a construction of period n reads at the root angle $2\pi j/n$, that angle reduces to $2\pi/m$ with $m = n/\gcd(n, j)$, and the construction is the seed of period m expressed at index $n = km$. The radix then follows immediately:

$$n = km, \quad b = \sec^n \theta = (\sec^m \theta)^k = b_{\text{seed}}^k$$

This is an identity, not a phenomenon. It also corrects an error worth recording, because it was ours: an earlier form of this work applied a totient bound $\phi(n) \leq 4$ directly to n , treating every occurrence as primitive. That bound constrains m , not n . Applied to n it excludes $n = 16$ and $n = 24$, both of which demonstrably exist — they are the $m = 8$ seed at $k = 2$ and $k = 3$. The apparent counterexamples were never counterexamples to the seed theorem; they were lifts of a seed the bound had already admitted.

A consequence is that the period alone does not identify a construction. At $n = 24$ there are two distinct towers: the 60° seed at $k = 4$, with $b = 64^4 = 2^{24}$, and the 45° seed at $k = 3$, with $b = 16^3 = 2^{12}$. They share an index and nothing else. The address of a tower is the pair (m, k) .

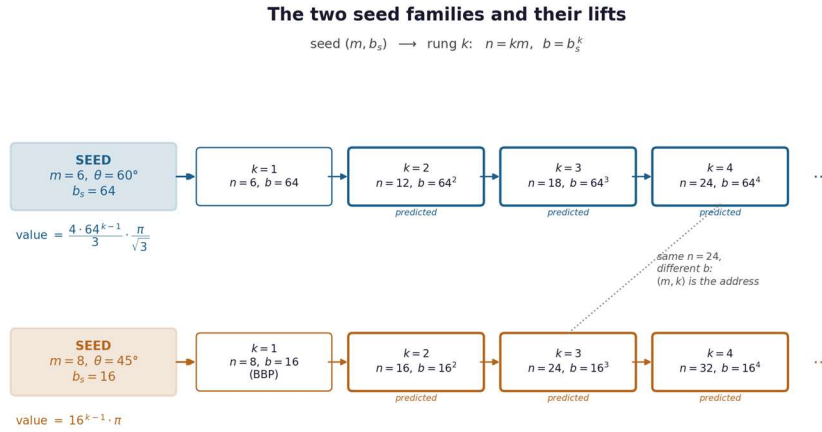


Figure 1. The two seed families and their lifts. A seed (m, b_{seed}) generates rung k with $n = km$ and $b = b_{\text{seed}}^k$. The classical base-16 formula for π is the $m = 8$ seed at $k = 1$. The rungs marked as predicted were constructed from the law and verified afterwards. The two families collide at $n = 24$ with different radices, which is why (m, k) rather than n is the address.

8. Where the coefficients sit

One element of the construction remained unexplained until last: why the two seeds place their weights on different shells. The base-64 seed uses shells $\{1, 2, 3, 4\}$; the base-16 seed uses $\{1, 4, 5, 6\}$. The rule turns out to be forced, and it is visible once the numerator is read as a difference operator.

Since $N(t) = c \cdot (1 - t) \cdot A(t)$, the coefficients of N are the successive differences of the coefficients of A . A shell is therefore empty exactly where two consecutive coefficients of A are equal. For $m = 6$, $A = t^2 + 2t + 4$ has ascending coefficients $[4, 2, 1]$ with no repetition, so no shell is empty and the support is consecutive. For $m = 8$, $A = (t^2 + 2)(t^2 + 2t + 2)$ has ascending coefficients $[4, 4, 4, 2, 1]$; the two repetitions annihilate shells 2 and 3, producing the familiar gaps.

The repetition count is itself determined. The annihilator has degree $m - 4$ — the m roots less the two real ones and the characteristic pair — and its descending coefficients are $[1, 2, 4]$ followed by $m - 6$ further constants, by the argument of §4. So there are exactly $m - 6$ empty shells, and the support is

$$\text{supp } N = \{1, m-4, m-3, m-2\}, \quad \text{weights } [4, -2, -1, -1].$$

For $m = 6$ this is $\{1, 2, 3, 4\}$ and for $m = 8$ it is $\{1, 4, 5, 6\}$. Combining the support with the radix gives a single closed generator for the whole class:

$$4S(1) - 2S(m-4) - S(m-3) - S(m-2), \quad b = \sec^m(2\pi/m)$$

Evaluated at $m = 8$ this is $4S(1) - 2S(4) - S(5) - S(6)$ over base 16 and period 8, which is the Bailey–Borwein–Plouffe formula, and it returns π to the precision tested. At $m = 6$ it returns $(4/3) \cdot (\pi/\sqrt{3})$. The classical formula is not a special arrangement that happens to work; it is one of the two instances the generator admits.

9. Prediction

Everything to this point explains objects already known, and a construction that only explains is weaker than one that anticipates. The lift law predicts specific unbuilt towers with specific coefficient vectors, so we built them.

At rung k the seed signature is replicated k times, on the seed's shell positions shifted by m each time, weighted by descending powers of b_{seed} . The prediction for $m = 6$, $k = 2$ — period 12, radix 4096 — was confirmed by integer-relation detection at 120 digits, returning the DC-blocked vector $(768, -384, -192, -192, 0, 0, 12, -6, -3, -3, 0, 0)$ with residual of order 10^{-119} . Its blocks are $192 \cdot [4, -2, -1, -1]$ and $3 \cdot [4, -2, -1, -1]$, a ratio of 64: the parent radix, appearing inside the child's coefficients. The $m = 8$, $k = 2$ prediction — period 16, radix 256 — likewise returned blocks $16 \cdot [4, -2, -1, -1]$ and $1 \cdot [4, -2, -1, -1]$ on shells $\{1,4,5,6\}$ and $\{9,12,13,14\}$, the parent's shell positions and their shift.

Higher rungs were then generated from the law and evaluated directly. All eight constructions return exact rational multiples of their family constant.

Family	k	n	b	value	status
45° (m=8)	1	8	16	$1 \cdot \pi$	classical (BBP)
45° (m=8)	2	16	256	$16 \cdot \pi$	predicted, verified
45° (m=8)	3	24	4096	$256 \cdot \pi$	predicted, verified
45° (m=8)	4	32	65536	$4096 \cdot \pi$	predicted, verified
60° (m=6)	1	6	64	$(4/3) \cdot \pi/\sqrt{3}$	prior work
60° (m=6)	2	12	4096	$(256/3) \cdot \pi/\sqrt{3}$	predicted, verified
60° (m=6)	3	18	262144	$(16384/3) \cdot \pi/\sqrt{3}$	predicted, verified
60° (m=6)	4	24	16777216	$(1048576/3) \cdot \pi/\sqrt{3}$	predicted, verified

The closed forms are $\pi \cdot 16^{(k-1)}$ for the 45° family and $(\pi/\sqrt{3}) \cdot 4 \cdot 64^{(k-1)}/3$ for the 60°. Six of the eight were not built before this work, and in each case the coefficient vector was written down from the law and the value computed afterwards.

10. Inverse recovery

A generator that cannot be run backwards is a recipe rather than a structure. Given only an observed pair (n, b) , the seed theorem leaves two possibilities, so the inverse is a finite check: for each seed $(m, b_{\text{seed}}) \in \{(6, 64), (8, 16)\}$, test whether m divides n and whether $b_{\text{seed}}^{(n/m)} = b$. Since both seeds are powers of two the test has a compact form — $\log_2 b = n$ identifies the 60° family and $\log_2 b = n/2$ the 45° — after which $k = n/m$ and $\theta = 2\pi/m$ follow.

Run on all eight constructions with only (n, b) supplied and no family indicated, the inverse recovers (m, k, θ, b_seed) in every case, and the $n = 24$ collision resolves into its two distinct towers. Feeding each recovered seed back through the generator of §8 reproduces the tower's value exactly at 80 digits. The loop closes in both directions.

The coefficient vector carries the same information independently, but reading it requires one more step than it first appears. At rung k the support is not the seed block but a union of translates,

$$\text{supp } N_{m,k} = \bigcup_{r=0}^{k-1} (\{1, m-4, m-3, m-2\} + rm),$$

so the largest occupied shell is $m-2 + (k-1)m = km - 2$. Reading the whole vector therefore gives n , not m — a distinction worth stating because an earlier form of this section conflated them, and the two coincide only at $k = 1$. The primitive period must instead be read from the block structure.

10.1 Uniqueness of the primitive block decomposition

This is enough to settle whether the decomposition is canonical, which had been the last gap in the construction. The seed theorem has already reduced the admissible primitive periods to two, and those two are distinguished by their second occupied shell.

Theorem. *The support of an admissible $N_{m,k}$ determines (m, k) uniquely.*

Proof. Every block begins with shell 1 and, for $m = 6$, ends at 4 with the next block beginning at 7; for $m = 8$ it ends at 6 with the next beginning at 9. Blocks therefore never overlap, and the second-smallest occupied shell always lies in the first block. That shell is $m-4$, which is 2 when $m = 6$ and 4 when $m = 8$. Since the seed theorem admits no other primitive period, m is determined. The largest occupied shell is $km - 2$, giving n , and hence $k = n/m$. ■

The consequence is that an observed tower carries its own address in three independent ways: numerically in the pair (n, b) , geometrically in the support, and by amplitude in the ratio between adjacent coefficient blocks, which is the parent radix. These need not all be used — the support argument alone establishes uniqueness once the seed theorem is granted — but their agreement is a cross-check the construction did not have to provide and does. A useful illustration is the $n = 24$ collision of §7: the two towers sharing that index have the same largest shell and are separated by their second, which is 2 for the 60° family and 4 for the 45° . Over the first six rungs of both families there are no coincident supports.

With uniqueness in hand the classified family is a bijection between tower addresses and admissible coefficient constructions: the forward generator produces $N_{m,k}$ from (m, k) , and the inverse recovers (m, k) from $N_{m,k}$ with no ambiguity.

11. The same reading in other substrates

The result above is a theorem about one family, and the closure language could be dismissed as vocabulary laid over ordinary linear algebra. Three further systems, chosen to share no mathematics with the first or with

each other, argue otherwise. In each, C_0 and C_1 are read off the system's own construction, the closure is written explicitly, and the admissible set is counted with each channel removed.

	Coefficient space	SHA-256 round	Duffing oscillator	Quantum spectrum
C_0	pole geometry	round law (Ch, Maj, Σ , K)	autonomous law	operator H
C_1	zero-sum	message drive W_t	forcing $y\cos\omega t$	admissibility
$\mathcal{M} = 0$	$\Phi c = 0$	$R(S_t, W_t) - S_{t+1} = 0$	$P(x, \dot{x}) - (x, \dot{x}) = 0$	$(H-E)\psi = 0, \psi(\infty) = 0$
C_0 alone	2-dimensional	2^{32} values	equilibria only	every E admits a solution
C_1 alone	5-dimensional	2^{32} values	1-dim continuum	no operator to satisfy
both	1-dimensional	unique	isolated orbits	isolated eigenvalues
observable	$9 \cdot [4, -2, -1, -1]$	the drive word	phase-locked orbit	the spectrum

Three of these are worth a sentence each. In the SHA-256 round the closure solves exactly for the drive word in 20000 of 20000 random trials, while either channel alone leaves 2^{32} admissible values; this also locates the function's one-wayness as a change of medium at the feed-forward rather than a destruction of information. In the driven Duffing oscillator the drive alone admits a one-dimensional continuum of solutions — integrating $\ddot{x} = y\cos\omega t$ over a period gives $\dot{x}(T) = \dot{x}(0)$ identically and $x(T) - x(0) = \dot{x}(0)T$, so the closure requires only $\dot{x}(0) = 0$ — while both channels give isolated phase-locked orbits whose stability changes between drive amplitudes 0.25 and 0.30. Here the collapse is weaker than elsewhere and we say so: the law alone does have isolated solutions, but they are static equilibria, not orbits.

The quantum case is the sharpest and also the most familiar, which is the point. For the harmonic oscillator the differential equation alone admits a solution at every energy — integrating outward from the origin at $E = 0.30, 0.70, 1.00$ and so on returns perfectly good solutions, which happen to diverge. Imposing decay at infinity leaves $E = 1/2, 3/2, 5/2, \dots$ and the solutions at those energies fall by twelve orders of magnitude relative to their neighbours. The implemented residual is $R(E) = |\psi_E(x_{\max})|$ from an outward integration with $\psi(0) = 1, \psi'(0) = 0$ and $x_{\max} = 6$; diagonalisation of the same operator returns the same eigenvalues to 10^{-5} . The statement is not that boundary conditions are mysterious; it is that a spectrum is the joint zero set of an operator equation and an admissibility condition, which is the architecture of §2 in the substrate where physics already knew it.

12. On arriving from the constraint side

Little in this paper is unowned. The Bailey–Borwein–Plouffe formula is famous. The degree of $2\cos(2\pi k/n)$ over the rationals is classical, as is the abelian character of cyclotomic fields and the irrationality machinery in the neighbourhood of Niven's theorem. The quantisation argument of §11 is textbook. We have not located a prior statement of the classification in §6 or of the generator in §8, but that is a statement about our search and not about the literature, and we would want it checked properly before any priority is claimed.

What we take to be the result is the direction of arrival. These objects were not assembled from their known properties and then re-described; they were reached by imposing constraints and reading what survived. The signature $[4, -2, -1, -1]$ was not put in. The radices 64 and 16 were outputs of $\text{Re}(z_{\text{char}}) = 1$. The support $\{1, m-4, m-3, m-2\}$ came from the annihilator's degree. And the classification came from an obstruction in field theory that the construction ran into rather than invoked. When a constraint-first route reaches a known landmark, the agreement is evidence that the route is real; when it then keeps going and produces towers nobody had built, the route is doing work the landmark cannot do.

This is also why the negative controls matter more than the successes. A method that arrives at known mathematics from an unusual direction is only interesting if it can also fail, and §2.3 shows the four ways it does.

13. What remains

The classified branch is closed. Seeds, generator, lifts, support, prediction, inverse and uniqueness form a forward–inverse pair with no free parameter left in it. What remains is of three kinds.

The first is a matter of generality. The seed theorem is stated for the primitive angle $2\pi/m$ and reaches the general root angle $2\pi j/m$ by reduction; a direct treatment would be cleaner and might extend to radices that are integers without being powers of two, a case the present argument passes over because both surviving seeds happen to be powers of two.

The second is the closure architecture itself. Section 11 exhibits it in four substrates and Section 2.3 shows four ways it fails, but exhibition is not characterisation: we cannot yet say which systems must admit such a reading, nor is the inverse established outside the two cases where it was run. Recovering the drive word from a Boolean round is demonstrated; recovering an operator and its admissibility condition from a spectrum is a genuinely harder inverse problem, and we make no claim about it here.

The third is not mathematics but diligence. The claim that this classification and this generator are unstated in the literature rests on our own searching, which is not the same as a search. That should be done properly before any of it is described as new, and the surrounding machinery — cyclotomic degree theory, the abelian character of these fields, the classical digit-extraction formula itself — is emphatically not ours.

The method for closing what is left is the one used throughout: state the two conditions, impose them at once, and read what neither can move.

$$\mathcal{M}(C_0, C_1; \mathbf{x}) = 0$$

Computational note. Every numerical statement was produced by executed code before being written. The rank-5 operator for $64 - t^6$ and its kernel $(16, -24, -8, -6, 1, 0)$ with $\Sigma z = -21$; the admissible dimensions $6/2/5/1$ and the strict independence of the constraints; the four negative controls; the ledger identity $\sqrt{3} = (64/16)(9/4)(\sqrt{3}/9)$; the base-forcing $\sec^6(60^\circ) = 64$ and

sec⁸(45°) = 16; the seed scan over $m = 3 \dots 4000$; the integer-relation detection for the $m = 6, k = 2$ and $m = 8, k = 2$ towers at 120 and 110 digits with residuals of order 10^{-119} and 10^{-110} ; the eight tower evaluations and their exact rational ratios; the inverse recovery and 80-digit round-trip on all eight; the SHA-256 drive-word recovery at 20000/20000; the Duffing solution-set analysis and Poincaré eigenvalues at $\gamma = 0.15 \dots 0.35$; and the harmonic-oscillator shooting and diagonalisation. Python with mpmath, sympy, numpy and scipy at 40–120 decimal digits as precision required. Source accompanies this paper.